

SET 1(NEW MANUAL)

Experiment 1: Smart Home Light Control System

Question:

Design and simulate a Smart Home Light Control System using Arduino in Tinkercad to control household lighting through digital input/output operations. Demonstrate ON/OFF automation of lights using appropriate electronic components and analyze the working of the system.

Experiment 2: Temperature Monitoring System

Question:

Develop and simulate a Temperature Monitoring System using Arduino and temperature sensors in Tinkercad to measure and display real-time temperature data. Analyze the sensor readings and demonstrate environmental monitoring functionality.

Experiment 3: Distance Alert System

Question:

Design and implement a Distance Alert System using an ultrasonic sensor and Arduino in Tinkercad to detect nearby obstacles and generate alert notifications based on predefined distance thresholds.

Experiment 4: Home Automation System

Question:

Create and simulate a Home Automation System using Arduino in Tinkercad for controlling multiple home appliances electronically. Demonstrate automation features and explain the operational workflow of the system.

SET 2(FROM OBSERVATION & RECORD)

Experiment 1: LED Control Using Push Button

Question:

Design and simulate an LED control circuit using a push button and Arduino in Tinkercad to demonstrate digital input and output operations.

Experiment 2: RGB LED Interfacing with Arduino

Question:

Develop and simulate an RGB LED interfacing circuit using Arduino in Tinkercad to generate multiple colors using PWM-based control techniques.

Experiment 3: LED Blink Rate Control Using Potentiometer

Question:

Design and implement a potentiometer-based LED brightness and blink rate control system using Arduino in Tinkercad to analyze analog input processing.

Experiment 4: Light Detection Using Photo Resistor**Question:**

Create and simulate a light detection system using a photo resistor (LDR) and Arduino in Tinkercad to detect variations in ambient light intensity and control LED operation.

Experiment 5: Servo Motor Interfacing with Arduino**Question:**

Design and simulate a servo motor control system using Arduino in Tinkercad to demonstrate angular position control using pulse width modulation techniques.

Experiment 6: Active Buzzer Interfacing with Arduino**Question:**

Develop and simulate an active buzzer interfacing circuit using Arduino in Tinkercad to generate different audio tones and alert signals.

Experiment 7: Relay Interfacing with Arduino**Question:**

Design and implement a relay control system using Arduino in Tinkercad to demonstrate switching operations for controlling external electronic devices.

Experiment 8: Intrusion Detection System Using Ultrasonic Sensor**Question:**

Create and simulate an intrusion detection system using an ultrasonic sensor, LEDs, buzzer, and Arduino in Tinkercad to detect object movement and generate security alerts.

Experiment 9: Directional Control of DC Motor Using Arduino**Question:**

Design and simulate a directional control system for a DC motor using Arduino in Tinkercad to demonstrate forward and reverse motor rotation control techniques.

S.NO	REG.NO	NAME	Experiment Numbers
1.	23TK0001	ABHINAYAA S	Set 1: Exp 1 & Set 2: Exp 3
2.	23TK0002	ANGEL RANI A	Set 1: Exp 3 & Set 2: Exp 5
3.	23TK0003	ASHWIN A	Set 1: Exp 2 & Set 2: Exp 1
4.	23TK0004	BALAMURALIKRISHNA S	Set 1: Exp 4 & Set 2: Exp 6
5.	23TK0005	BHARATHI M	Set 1: Exp 1 & Set 2: Exp 3
6.	23TK0006	CHARUKESH S	Set 1: Exp 2 & Set 2: Exp 4
7.	23TK0008	DAYANITHI V	Set 1: Exp 3 & Set 2: Exp 5
8.	23TK0009	DEEPIKA B	Set 1: Exp 4 & Set 2: Exp 6
9.	23TK0010	GOVINDARAJ A	Set 1: Exp 2 & Set 2: Exp 1
10.	23TK0011	HARISH S	Set 1: Exp 1 & Set 2: Exp 2
11.	23TK0012	HARSHINI B	Set 1: Exp 2 & Set 2: Exp 8
12.	23TK0013	JAGANATHAN M	Set 1: Exp 3 & Set 2: Exp 9
13.	23TK0014	KALAI SELVAN I	Set 1: Exp 4 & Set 2: Exp 5
14.	23TK0015	KARTHIK B	Set 1: Exp 2 & Set 2: Exp 1
15.	23TK0016	KARUNYA R	Set 1: Exp 1 & Set 2: Exp 7
16.	23TK0017	KEERTHANA G	Set 1: Exp 3 & Set 2: Exp 8
17.	23TK0018	KISHORE KUMAR R	Set 1: Exp 2 & Set 2: Exp 1
18.	23TK0019	LAKSHMIPRIYA V	Set 1: Exp 4 & Set 2: Exp 6
19.	23TK0020	MADHANKUMAR L	Set 1: Exp 1 & Set 2: Exp 2
20.	23TK0021	MOHAMED FAWAZ A	Set 1: Exp 3 & Set 2: Exp 5
21.	23TK0022	MOHAMED THANVEER I	Set 1: Exp 2 & Set 2: Exp 1
22.	23TK0023	MOHANAKUMAR S	Set 1: Exp 4 & Set 2: Exp 6
23.	23TK0024	MUKESH M K	Set 1: Exp 1 & Set 2: Exp 7
24.	23TK0025	MUKILAN R	Set 1: Exp 2 & Set 2: Exp 4
25.	23TK0026	NAREN KUMAR P	Set 1: Exp 3 & Set 2: Exp 5
26.	23TK0027	NIDHESHWAR P	Set 1: Exp 4 & Set 2: Exp 6
27.	23TK0028	PAVITHRA P	Set 1: Exp 1 & Set 2: Exp 7
28.	23TK0029	PRIYADARSHAN S	Set 1: Exp 2 & Set 2: Exp 1

29.	23TK0030	RAGHAVI SREE S	Set 1: Exp 3 & Set 2: Exp 9
30.	23TK0031	RITHIKKA V T	Set 1: Exp 2 & Set 2: Exp 8
31.	23TK0032	S DHINESH KUMAR	Set 1: Exp 4 & Set 2: Exp 5
32.	23TK0033	SAMHITA V	Set 1: Exp 1 & Set 2: Exp 3
33.	23TK0034	SANNIYASI S	Set 1: Exp 2 & Set 2: Exp 1
34.	23TK0035	SHIVENDRAN P	Set 1: Exp 3 & Set 2: Exp 5
35.	23TK0036	SHYAM SUNDAR K	Set 1: Exp 1 & Set 2: Exp 2
36.	23TK0037	SIVAGNANAM R	Set 1: Exp 4 & Set 2: Exp 6
37.	23TK0038	SOMASUNDARAM C	Set 1: Exp 1 & Set 2: Exp 7
38.	23TK0039	SUBALAKSHMI K	Set 1: Exp 2 & Set 2: Exp 8
39.	23TK0040	SURENDAR KHADAN C	Set 1: Exp 4 & Set 2: Exp 5
40.	23TK0041	SYED RIASUDDIN S S	Set 1: Exp 3 & Set 2: Exp 9
41.	23TK0042	VAISHNAVI E	Set 1: Exp 2 & Set 2: Exp 8
42.	23TK0043	VEDHANTH R	Set 1: Exp 4 & Set 2: Exp 6
43.	23TK0044	VISHNU M	Set 1: Exp 3 & Set 2: Exp 5
44.	23TKL00 1	MOHANA. B	Set 1: Exp 1 & Set 2: Exp 3